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CHE 313 Transport Phenomena III

Exam #1 (2/19/2020)

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We are separating acetone and ethanol in a distillation column with a total condenser and a partial reboiler. The column has two feeds and operates at a pressure of 1.0 atm. Feed 1 is a saturated liquid that is 60.0 mol% acetone and has a feed rate of F1 = 200.0 kmol/hr. The second feed is 40.0 mol% acetone, has a feed rate of F2 = 300.0 kmol/hr, and is a two-phase mixture that is 20% vapor. Assume CMO is valid. Bottoms product is 10.0 mol% acetone, and distillate product is 80.0 mol% acetone. External reflux ratio L/D = 2.0. XB

XD

1) (5 pts) Draw a schematic diagram of the distillation column and label all known variables (use V & L for enriching section, V' & L' for middle section, and $\bar{L}$ & $\bar{V}$ for stripping section)
2) (10 pts) Draw mass balance envelopes for each section (total four balance envelopes)
3) (15 pts) Find D, B, L, V, V', L', $\bar{L}$, and $\bar{V}$ (hint: set up overall mass balance and component mass balance)
4) (5 pts) Find slope, y-intercept, and coordinate on y=x curve of the top operating line: intersection

$$y = \left(\frac{L}{V}\right)x + \left(1 - \frac{L}{V}\right)x_D$$

5) (5 pts) Find slope, x-intercept, and coordinate on y=x curve of the bottom operating line:

$$y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$$

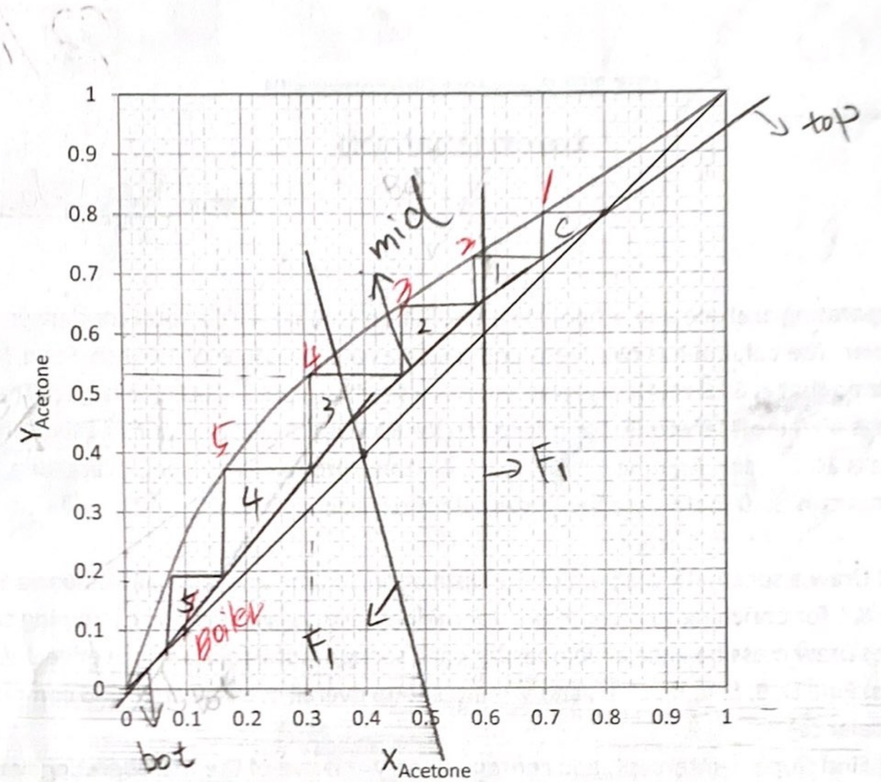
6) (10 pts) The equation for the feed line is:

$$y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1 - q_i}$$

$$q_i = \frac{L_{Fi}}{F_i}$$

Find $\bar{q}_1$, $\bar{q}_2$ coordinates on y=x line, and slopes of two feed lines (F1 & F2), and x-intercept of F2.

7) (5 pts) Write overall mass balance and component mass balance around the top column (hint: mass balance between top feed and distillate)
8) (15 pts) Derive an operating equation for the middle section. Find slope and y-intercept
9) (10 pts) Draw the top operating, bottom operating, two feeds, and middle operating lines



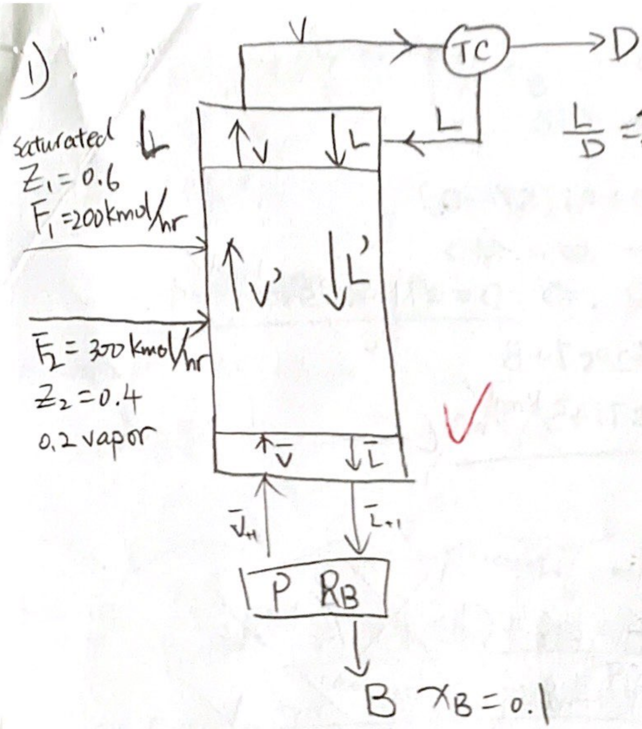
10) (10 pts) Find the number of stages and the optimum locations of feed 1 and feed 2 stages

11) (10 pts) Find mole fraction (x_3, y_3) of the stage 3

Fu Wu

CMO

1)

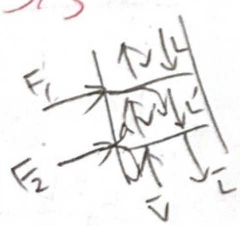


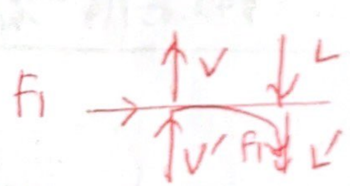
$x_D = 0.8$

$\frac{L}{D} = 2$

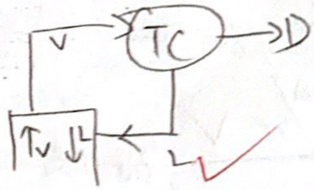
$x_B = 0.1$

✓ 5/5





2) Top:



✓

$$V = L + D$$

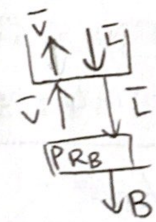
$$Vy = Lx + Dx_D$$

$$y = \frac{L}{V}x + \frac{D}{V}x_D$$

$$y = \frac{L}{V}x + \frac{V-L}{V}x_D$$

$$y = \frac{L}{V}x + \left(1 - \frac{L}{V}\right)x_D$$

Bot:



✓

$$\bar{L} = \bar{V} + B$$

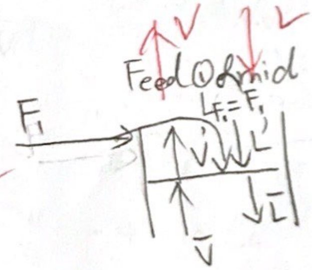
$$\bar{L}x = \bar{V}y + Bx_B$$

$$y = \frac{\bar{L}x - Bx_B}{\bar{V}}$$

$$y = \frac{\bar{L}}{\bar{V}}x - \left(\frac{\bar{L} - \bar{V}}{\bar{V}}\right)x_B$$

$$y = \frac{\bar{L}}{\bar{V}}x + \left(1 - \frac{\bar{L}}{\bar{V}}\right)x_B$$

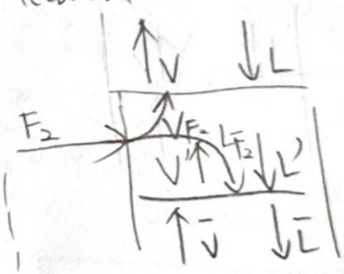
Feed ① mid



$$V' + \bar{L} = L' + L_F + \bar{V}$$

-1

Feed ② mid



$\frac{V_F}{F} = 0.2$ ✓

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3) $F_1 + F_2 = D + B$

$z_1F_1 + z_2F_2 = x_D D + x_B B$

$200 + 300 = D + B$

$0.6(200) + 0.4(300) = 0.8D + 0.1B$

$500 = D + B$ $\quad B = 500 - D$

$240 = 0.8D + 0.1B$

$240 = 0.8D + 0.1(500 - D)$

$240 = 0.8D + 50 - 0.1D$

$190 = 0.7D \Rightarrow D = 271.42857$ kmol/hr ✓

$500 = 271.42857 + B$

$\Rightarrow B = 228.57143$ kmol/hr ✓

$\frac{L}{D} = 2$

$L = 2D = 2 \times 271.42857$

$L = 542.85714$ kmol/hr ✓

$V = L + D$

$V = 542.85714 + 271.42857$

$V = 814.28571$ kmol/hr $= V'$ ✓

b/c CMO

$\cancel{L' = L + L_{F_1} + L_{F_2} = 982.85714 \text{ kmol/hr}}$ (542.867, 200, 0.8×300)

$V' = \bar{V} + V_{F_2} = 814.28571$ kmol/hr ✓

$\bar{V} + V_{F_2} = V$ $\quad V_{F_2} = 0.2F_2 = 60$ kmol/hr

$\bar{V} = V - V_{F_2}$

$= 814.28571$ kmol/hr $- 60$

$\bar{V} = 754.28571$ kmol/hr ✓

$\bar{L} = L + L_{F_1} + L_{F_2}$ $\quad L_{F_1} = 1 \times F_1 = 200$ kmol/hr

$L_{F_2} = 0.8F_2 = 240$ kmol/hr

$\bar{L} = 542.85714 + 200 + 240$

$\bar{L} = 982.857$ kmol/hr ✓

sat. liq.

$L' = L + F_1$

$= 742.8$ kmol/hr

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4) $y = \frac{L}{V}x + (1 - \frac{L}{V})x_D$

$V = L + D$ $\quad \frac{L}{D} = 2$

$L = 2D$ or $D = \frac{L}{2}$

slope: $\frac{L}{V} = \frac{2}{3}$ ✓

$V = L + \frac{L}{2}$

$V = \frac{3}{2}L$

$\frac{L}{V} = \frac{2}{3}$

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y-int $= (1 - \frac{2}{3})(0.8)$

$= \frac{1}{3}(0.8)$ ✓

y-int $= 0.26667$ ✓

(0, 26667)

$y = \frac{2}{3}x + 0.2667$

— coordinate on y=x curve of top op

$y = x = x_D$

(0.8, 0.8) ✓

5) $y = \left(\frac{\bar{L}}{\bar{V}}\right)x - \left(\frac{\bar{L}}{\bar{V}} - 1\right)x_B$

$\frac{\bar{L}}{\bar{V}} = \frac{982.857}{754.28571} = 1.30$

slope: $\frac{\bar{L}}{\bar{V}} = 1.30$

$y = 1.30x - 0.03$

x-int: 0.23

(0.023)

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-coordinate on $y=x$ curve of Bot op

$y = x = x_B$

$(0.1, 0.1)$

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6) $y = \frac{q_i}{q_i - 1}x + \frac{z_i}{1-q_i}$ $z_1 = 0.6$

for feed ① → saturated liquid

$q = 1$

$(q-1)y = qx - z_1$

$0 = 1x - z_1$

$x = z_1 = 0$

Feed 1 line: $x = z_1 = 0.6$

-slope: ∞ → vertical line

- x-int: $x = z_1 = 0.6$

$(0.6, 0)$

- intersection on $y=x$ line

$(0.6, 0.6)$

$z_2 = 0.4$

for feed ②

$q = \frac{(1-0.2)F_2}{F_2} = \frac{0.8}{1} = 0.8$

$y = \frac{0.8}{0.8-1}x + \frac{0.4}{1-0.8}$

$y = -4x + 2$

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- slope: $\frac{q}{q-1} = -4$

- x int: $0 = -4x + 2$ $x = \frac{1}{2}$

$(\frac{1}{2}, 0)$

- intersection on $y=x$ line

$(0.4, 0.4)$

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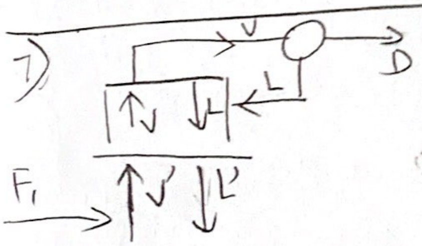
7)



$F + V' = L' + D$

$F_1 z_1 + V'y = L'x + Dx_D$

5/5

8) $F_1 z_1 + V'y = L'x + D\cdot x_D$ $\quad L' =$

$$y = \frac{L'x + Dx_D - F_1 z_1}{V'}$$

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$$y = \frac{L'}{V'}x + \frac{Dx_D - F_1 z_1}{V'}$$ ✓

$$= \frac{\cancel{982.85714}\ 742.8}{814.28571}x + \frac{0.8\times 271.43 - 200\times 0.6}{814.28571}$$

$y = \cancel{1.207}\ 0.912\, x + 0.1193$

-1

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9) draw 9/10

-1

10) optimum location | total # of stage : 5 (6-1) b/c total condenser

feed ① $F_1 \longrightarrow$ ~~1st~~ 2nd stage

feed ② $F_2 \longrightarrow$ ~~3rd~~ 4 stage

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11) $x_3 \cong$ ~~0.31~~ 0.49

$y_3 \cong$ ~~0.521~~ 0.67

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-2

Should find for x_4, y_4 in your case!